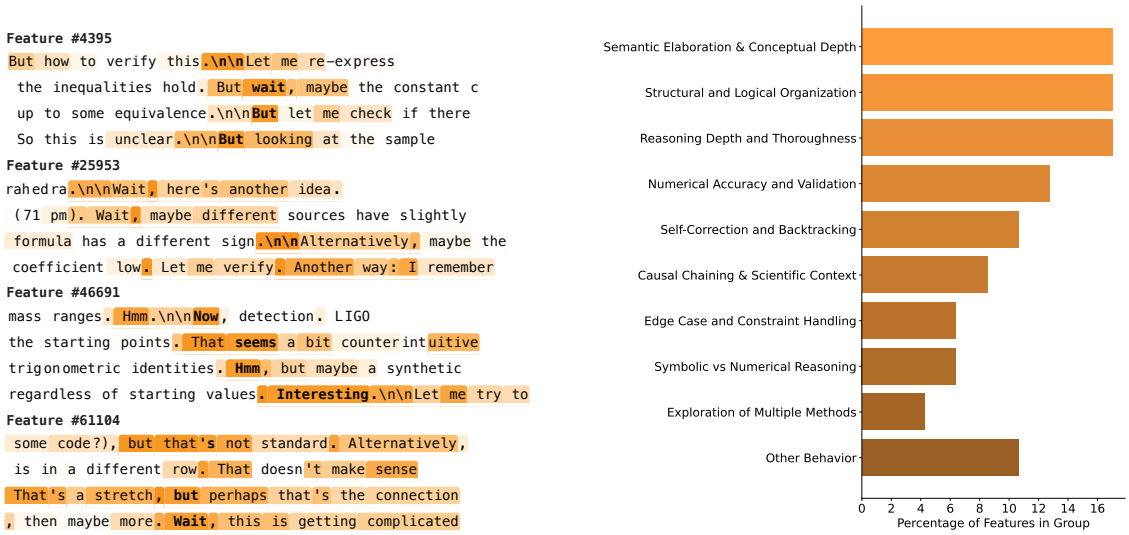

(a) Top-activating examples from the manually verified set of features.

(b) Distribution of manually verified set of features on function groups generated by GPT-4o.

Figure 3: Interpretability results for manually verified set of features in our SAE: (a) Examples of feature interfaces used in manual interpretation experiments, (b) Distribution of reasoning features on function groups obtained by automatic interpretation pipeline by using GPT-4o as a judge.

of i-th feature to reasoning. Using a dataset of model's activations (see details in Sec. 4.1) $\mathcal{D} = \mathcal{D}_{\mathcal{R}} \cup \mathcal{D}_{\neg\mathcal{R}}$, where $\mathcal{D}_{\mathcal{R}}$ contains token activations corresponding to words in $\mathcal{R}$ and $\mathcal{D}_{\neg\mathcal{R}}$ contains all other activations, we first define a score:

$$s_i = \frac{\mu(i, \mathcal{D}_{\mathcal{R}})}{\sum_j \mu(j, \mathcal{D}_{\mathcal{R}})} - \frac{\mu(i, \mathcal{D}_{\neg\mathcal{R}})}{\sum_j \mu(j, \mathcal{D}_{\neg\mathcal{R}})}, \quad (3)$$

where $\mu(i, \mathcal{D}) = \frac{1}{|\mathcal{D}|} \sum_{x \in \mathcal{D}} f_i(x)$ is the average activation value of the i-th feature on dataset $\mathcal{D}$. This score is similar to the one in (Cywiński and Deja, 2025) and identifies features that concentrate the most of their activation mass on reasoning words.

However, analysis of feature activations only on individual words may miss important contextual information. The words in $\mathcal{R}$ are critical indicators of the reasoning process and also serve as transition points, signaling shifts in the thought process, uncertainty, or reflection. Therefore, a feature involved in reasoning should activate not only on the reasoning words, but also as the model approaches and continues through these transitions. To capture it, we define $\mathcal{D}^{\mathrm{W}}_{\mathcal{R}}$ as the dataset that contains activations within a fixed-width context window around tokens corresponding to words in $\mathcal{R}$, and $\mathcal{D}^{\mathrm{W}}_{\neg\mathcal{R}}$ contains all other activations. We modify Eq. 3 to use the new version of the datasets.

To penalize features that activate only on a small fraction of $\mathcal{R}$, we further introduce an *entropy penalty*. For i-th feature, we first calculate $\mu(i, \mathcal{D}^{\mathrm{W}}_{r_j})$ for each word $r_j \in \mathcal{R}$, normalize these values into a probability distribution $p_i(r_j) = \frac{\mu(i, \mathcal{D}^{\mathrm{W}}_{r_j})}{\sum_{k \in \mathcal{R}} \mu(i, \mathcal{D}^{\mathrm{W}}_{r_k})}$, and compute the entropy:

$$\mathrm{H}_i = -\frac{1}{\log |\mathcal{R}|} \cdot \sum_j^{|\mathcal{R}|} p_i(r_j) \log p_i(r_j). \quad (4)$$

Here, $\log |\mathcal{R}|$ normalizes the entropy to $[0, 1]$, with $\mathrm{H}_i = 1$ indicating perfect uniformity over $\mathcal{R}$. By adding the entropy penalty in Eq. (3), we define the `ReasonScore` for the i-th SAE feature as:

$$\begin{aligned} \texttt{ReasonScore}_i = & \frac{\mu(i, \mathcal{D}^{\mathrm{W}}_{\mathcal{R}})}{\sum_j \mu(j, \mathcal{D}^{\mathrm{W}}_{\mathcal{R}})} \cdot \mathrm{H}_i^{\alpha} \\ & - \frac{\mu(i, \mathcal{D}^{\mathrm{W}}_{\neg\mathcal{R}})}{\sum_j \mu(j, \mathcal{D}^{\mathrm{W}}_{\neg\mathcal{R}})}. \end{aligned} \quad (5)$$

where α controls the trade-off between specificity ($\alpha \to 0$) and generalization ($\alpha > 1$).

We identify the set of reasoning features in a SAE based on their `ReasonScore` and define the corresponding set of feature indices as:

$$\mathcal{F}_{\mathcal{R}} = \{i \mid i \in [1, m],\ \texttt{ReasonScore}_i > \tau\}, \quad (6)$$

where τ is the q-th quantile of the `ReasonScore` distribution across all features.

4 Evaluation

In this section, we analyze how effectively our discovered features model reflection, uncertainty, and exploration within the *reasoning* model. We discuss our experimental setup (Sec. 4.1), perform manual and automatic interpretation of the features we find (Sec. 4.2), and conduct steering experiments with these features on various benchmarks (Sec. 4.3). Finally, we apply the model diffing technique to demonstrate that these features exist only in models with reasoning capabilities (Sec. 4.4).

4.1 Experimental Setup

Model. We apply SAE to the output activations from the 19-th layer of the DEEPSEEK-R1-LLAMA-8B model. This model was selected for its *reasoning* capabilities and open-source availability. The 19-th layer ($\approx 60\%$ model depth) was chosen because at this point LLMs predominately store the most of their knowledge (Chen et al., 2023; Jin et al., 2024). We provide results for other layers of DEEPSEEK-R1-LLAMA-8B and another model family in Appx. D.

Data. We train SAE on the activations of the model generated using text data from the LMSYS-CHAT-1M (Zheng et al., 2023) and OPENTHOUGHTS-114K (OpenThoughts, 2025) datasets. The first provides a broad and diverse spectrum of real-world text data, which we denote as *base data*, while the latter provides high-quality reasoning traces generated by DEEPSEEK-R1 for math, science, code, and puzzle samples, which we denote as *reasoning data*. The SAE is trained on 1B tokens, evenly split between the two datasets, with a context window of 1,024 tokens.

Training. We set the SAE dictionary dimensionality to $m = 65{,}536$, which is 16 times larger than the model activation size $n = 4{,}096$ following established practices (Lieberum et al., 2024), and adopt the same training settings as in the Anthropic April update (Conerly et al., 2024). We train with the Adam optimizer (Kingma and Ba, 2014) with $(\beta_1,\ \beta_2) = (0.9, 0.999)$, batch size of 4,096, and a learning rate $\eta = 5 \times 10^{-5}$. The learning rate is decayed linearly to zero over the last 20% of training. The gradient norm is clipped to 1. We use a linear warmup for the sparsity coefficient from $\lambda = 0$ to $\lambda = 5$ over the first 5% training steps.

Evaluation. We use the mean L0-norm of latent activations, $\mathbb{E}_x \|f(x)\|_0$, as a measure of sparsity.

Feature #	AIME 2024		MATH-500		GPQA Diamond	
	maj@4	tokens (K)	maj@4	tokens (K)	maj@4	tokens (K)
No steering	53.3	12.4	93.2	3.9	50.0	7.9
3942	56.7	11.1	93.0	3.4	46.5	6.7
4395	56.7	**14.7**	**95.4**	4.1	52.0	8.5
16441	60.0	14.0	95.0	4.1	54.0	8.3
16778	56.7	14.1	94.0	**4.7**	51.0	**9.0**
25953	60.0	12.8	94.2	4.2	53.0	8.1
46691	56.7	14.0	94.2	4.2	**54.0**	8.0
61104	**66.7**	12.0	95.0	3.6	53.0	7.5

Table 1: Performance and average number of output tokens for different steering experiments on reasoning-related benchmarks.

To measure reconstruction quality, we use fraction of variance of the input explained by the reconstruction. Both metrics were computed on 2,048 sequences of length 1,024.

At a L0 of 86 the reconstruction of our SAE explains 68.5% of the variance in model activations. This shows that our SAE achieves reliable reconstruction performance at a low sparsity level, allowing a decomposition of raw activations into interpretable features.

`ReasonScore`. We calculate `ReasonScore` (Eq. 5) on 10M tokens from the OPENTHOUGHTS-114K dataset. To collect $\mathcal{D}^{\mathrm{W}}_{\mathcal{R}}$, we use an asymmetric window with 2 preceding and 3 subsequent tokens, following established practices in Keyphrase extraction (Mihalcea and Tarau, 2004; Breidt, 1996; Zhang et al., 2020) We set $\alpha = 0.7$ for the *entropy penalty* as a reasonable default. Based on the empirical analysis of `ReasonScore` distribution (see Appx. A.3), we set $q = 0.997$ in Eq. (6), resulting in $|\mathcal{F}_{\mathcal{R}}| = 200$ features.

4.2 Interpretability of Reasoning Features

Manual Interpretation. To evaluate the features we found with `ReasonScore`, we manually interpret each feature from $\mathcal{F}_{\mathcal{R}}$ (200 in total). For each feature, we find the examples in a subset of the OPENTHOUGHTS-114K corpus that caused the feature to activate, and construct the interface proposed in (Bricken et al., 2023). This mainly includes examples of when the feature activates, its effect on the logits when it does, and other statistics. We determine whether a feature qualifies as a good reasoning candidate if: (1) when it is active, the relevant concept is reliably present in the context, (2) it triggers in various examples of reasoning tasks, and (3) its activation impacts interpretable logits that correspond to reasoning processes.

Through our analysis, we identify three behavioral modes that characterize models' reasoning process:

- **Uncertainty**: Moments where the model exhibits hesitation, doubts, and provisional thinking
- **Exploration**: Moments where the model considers multiple possibilities, connects ideas, examines different perspectives
- **Reflection**: Moments where the model revisits and reevaluates its previous steps

Our manual analysis reveals a set of 46 features that exhibit these patterns, which we believe are responsible for the reasoning mechanisms of the model. We denote this set by $\mathcal{F}_{\mathcal{R}}^{\text{manual}} \subset \mathcal{F}_{\mathcal{R}}$. In Fig. 3a, we provide examples of feature interfaces used for interpretation. The results demonstrate features that consistently activate in contexts representing model's uncertainty (#61104), exploration (#25953), and reflection (#4395, #46691). Additional examples of interfaces can be found in Appx. B.1.

Automatic Interpretation. To complement our manual analysis, we annotate these features with an automatic interpretation pipeline (Kuznetsov et al., 2025). This approach employs feature steering, a technique that modulates feature activations to analyze their functional influence. For each i-th feature, we estimate its maximum activation $f_i^{\max}$ using a subset of the OpenThoughts-114k corpus. During response generation, we modify model activations as follows:

$$x' = x + \gamma f_i^{\max} W_{\text{dec},i}, \tag{7}$$

where γ controls the steering strength.

To evaluate the impact of i-th feature on reasoning capabilities, we generate multiple outputs by varying $\gamma \in [-4, 4]$, pass them to GPT-4o, and ask it to generate an explanation or function that best describes the semantic influence caused by steering a feature. The result, shown in Fig. 3b, reveals that the features we found group into distinct reasoning-related patterns. Only a small fraction of features from $\mathcal{F}_{\mathcal{R}}^{\text{manual}}$ (5) was assigned to a mixed class "Other Behavior" containing mixed explanation. We provide a more comprehensive description of auto-interpretability pipeline results in Appx. B.2.

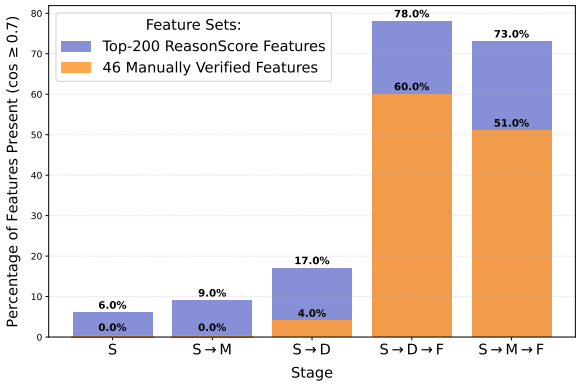

Figure 4: Percentage of ReasonScore features present at each stage of the diffing pipeline. The blue bars represent the features from $\mathcal{F}_{\mathcal{R}}$, the orange bars represent the $\mathcal{F}_{\mathcal{R}}^{\text{manual}}$ features. Features are considered present if their cosine similarity with any feature in corresponding stage's SAE is ≥ 0.7. Stages: (**S**) Base model + base data; (**S→D**) Base model + reasoning data; **S→M** Reasoning model + base data; (**S→D/M→F**) Reasoning model + reasoning data.

> **Takeaway 1**: Manual interpretation experiments confirm that ReasonScore identifies features that describe model's reasoning capabilities, revealing 46 features that represent uncertainty, exploration, and reflection. Automatic interpretation demonstrates that these features are causally linked to reasoning behavior.

4.3 Steering Reasoning Features

To demonstrate whether our interpretations of features describe their influence on model behavior, we further experiment with feature steering.

Our goal is to verify if steering reasoning features improve the LLM's performance on reasoning-related benchmarks. Following the setup in DeepSeek-R1, we evaluate performance on AIME 2024 (of America, 2024), MATH-500 (Hendrycks et al., 2021), and GPQA Diamond (Rein et al., 2023). To obtain steering results for i-th feature, we modify the activations during response generation according to Eq. (7). To determine the optimal steering strength that can influence model outputs without significantly damaging capabilities, we ran evaluations with a small subset of 10 reasoning features on MATH-500. We varied the steering strength γ from 1 to 8. Based on these experiments, we determined the optimal range $\gamma \in [1, 3]$, which aligns with the findings in (Durmus et al., 2024). For all subsequent experi-